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Brief Report

Using Time Trade-Off Values to Estimate EQ-5D-Y Value Sets: An Exploratory Study



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ABSTRACT

Objectives: The published international EQ-5D-Y-3L valuation protocol does not recommend the composite time trade-off (cTTO) method as the primary valuation method because of 2 major concerns. First, the cTTO method was shown to generate high values. Second, the cTTO method is not as feasible for valuing children's health as other established methods. This study aimed to explore the feasibility of using cTTO values alone to estimate EQ-5D-Y-3L value sets.

Methods: We analyzed the cTTO data derived from the recently completed Chinese EQ-5D-Y-3L valuation study in which a total of 28 health states were valued. We assessed the feasibility of the cTTO tasks in terms of survey completion time and participant-reported difficulty of understanding the task, differentiating the health states, and deciding the answer. We also examined the data distribution characteristics and modeled the data using different models.

Results: In total, 418 participants completed the cTTO interview. On average, participants took approximately 35.70 minutes (SD 12.42) to complete the interview and made 13.21 moves (SD 9.00) in the cTTO tasks. There were 74.16%, 59.33%, and 11.48% of participants indicated that it was easy to understand the cTTO tasks, easy to differentiate between the health states, and difficult to decide on an indifference point, respectively. The data distribution was smooth, and a random-effects model performed the best in terms of coefficient significance, monotonicity, and predictive accuracy.

Conclusion: Our finding suggests that estimating EQ-5D-Y-3L value sets using cTTO data alone is feasible and therefore could be considered as an option in future valuation studies for EQ-5D-Y-3L.

Keywords: composite time trade-off, EQ-5D-Y-3L, feasibility, value set.

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Introduction

Although remarkable progress in health state valuation has been made over the past decade,^{1,2} there is still further work to be done on the valuation of child health states and subsequent value set development. To address various methodological questions for valuing children's health, the EuroQol Group conducted a program of methodological research^{3–5} and developed the first international EQ-5D-Y-3L valuation protocol.⁶ This protocol aims to guide the development of value sets for child health states defined by EQ-5D-Y-3L.³ Several countries have since established their national value sets following this protocol.^{7–9}

The international EQ-5D-Y-3L valuation protocol recommends using discrete choice experiment (DCE) data as the primary preference data to estimate the relative importance of the 5 health dimensions and composite time trade-off (cTTO) data to anchor the estimated DCE values onto the quality-adjusted life-year (QALY) scale.⁶ This is in contrast to the protocol for valuing adult 5-level version of EQ-5D (EQ-5D-5L) health states, where cTTO is the recommended valuation method.^{1,10} The reason for downgrading the role of the cTTO method in the EQ-5D-Y-3L

valuation protocol was twofold. First, it is more feasible and cost-efficient to collect DCE data because it can be conducted online without assistance from trained interviewers. More importantly, a multicountry study found that adult respondents tended to avoid trading children's life for better health,³ indicating that cTTO values might not be able to identify the relative importance of the EQ-5D-Y-3L health dimensions.⁶ The authors of that study concluded that further investigation is needed to determine the suitability of using cTTO to value child health states.

Although high cTTO values were observed in several EQ-5D-Y-3L valuation studies,^{7,9,11} a Slovenian study⁸ observed negative mean cTTO values for 4 of 10 EQ-5D-Y-3L states, with the value for state “33333” (−0.691) being lower than all the EQ-5D-3L values for the country.¹² This finding challenges the assumption that the cTTO method might fail to identify the relative importance of the different health dimensions or levels of child health states. Nevertheless, most published EQ-5D-Y-3L studies, including the Slovenian study, valued only 10 states using the cTTO method, which does not allow for the estimation of a cTTO-only value set to assess this assumption.¹³

This study aimed to further explore the cTTO valuation of EQ-5D-Y-3L health states by using a cTTO design larger than the one used in published studies. We hypothesized that the cTTO method is feasible for valuing EQ-5D-Y-3L health states and that EQ-5D-Y-3L value sets can be estimated using cTTO values alone.

Methods

Our study used the cTTO data collected from the EQ-5D-Y-3L valuation study conducted in China,¹⁴ which followed a modified international EQ-5D-Y-3L valuation protocol.⁶ In that study, a total of 28 health states were valued using the cTTO method, which allows us to explore different ways of modeling the cTTO data. The study was approved by the Jinan University research ethics committee (approval number JNUKY-2021-002) and data collection was conducted between November 2019 and June 2020. The study design for the cTTO component is described below.

Health State Selection

The 28 EQ-5D-Y-3L health states selected for valuation included the 10 health states from the international EQ-5D-Y-3L valuation protocol. The remaining 18 health states were determined using an orthogonal design.¹³ The health states were divided into 3 blocks of 10 states, with state “33333” being included in every block. A sample of 400 respondents each valued one block of health states, which allows the estimation of a cTTO value set.^{13,15}

Sampling and Recruitment Strategy

A representative general Chinese population sample was recruited using a quota sampling method. Quotas for sex, age, education attainment, and *hukou* (rural or urban, a surrogate of socioeconomic status) were set based on the sixth National Population Census of China¹⁶ and applied to each of the 8 provinces or cities selected for participant recruitment, including Tianjin (North China), Shanghai, Jiangsu (East China), Guangdong (South China), Hubei (Central China), Guizhou, Chongqing, and Shanxi (West China). A quota of 50 participants was allocated to each recruitment city or province. A local team was assembled and tasked to be responsible for both participant recruitment and data collection. Participants were contacted and recruited from selected public locations or from the interviewers' social networks. Participants must (1) meet predetermined quotas, (2) be Chinese and have lived in China for at least 5 years before the interview, and (3) be well enough to provide an informed consent.

Data Collection Procedures

All participants were invited to a face-to-face, one-on-one computer-assisted personal interview. All interviews adhered to the EuroQol-Valuation Technology protocol.¹ After providing a brief introduction to the study, trained interviewers went through some warm-up questions, demonstration, and some practice of cTTO questions and then guided the participants to complete the valuation of 10 EQ-5D-Y-3L health states. In each cTTO task, participants were asked to imagine that an EQ-5D-Y-3L health state was experienced by a 10-year-old child and to state their preferences between 2 hypothetical lives (ie, life A and life B). Detailed descriptions of the cTTO elicitation procedures can be found in the EuroQol protocol for time trade-off valuation of health outcomes.¹⁰ Briefly, for the health states that a participant considered to be “better-than-dead,” the indifference point between life A (x years in full health followed by death) and life B (10 years in the health state being valued followed by death) was identified; for

“worse-than-dead” states, the indifference point between life A (x years in full health followed by death) and life B (10 years in full health followed by 10 years in the health state being valued and then death) was identified. The cTTO value was $x/10$ and $(x - 10)/10$ for the “better-than-dead” and “worse-than-dead” health states, respectively. Upon completion of the valuation tasks, participants were asked to report their experience of the valuation tasks by indicating their levels of agreement or disagreement for the following statements: (1) “I found it easy to understand the question,” (2) “I found it easy to tell the difference between the 2 health states,” and (3) “I found it difficult to make my decision.” Response options ranged from “agree” to “disagree.” Characteristics of the “10-year-old child” participants had in mind when completing the cTTO tasks were elicited. Demographics and health characteristics of the participants were also recorded.

In each study city or province, a graduate student from a local university was recruited and trained to be the interviewer. Following the EuroQol-Valuation Technology protocol, all 8 interviewers were trained in a 2-day workshop and their compliance to the interview protocol was monitored throughout the entire data collection period using a quality control (QC) procedure.¹⁷

Data Analysis and Model Evaluation

We assessed the feasibility of the cTTO tasks in terms of survey completion time and participant-reported difficulty in terms of (1) self-reported understanding of the task, (2) self-reported ease of differentiating the health states, and (3) self-reported difficulty in deciding on an indifference point when completing the cTTO task. We compared the results with those from the Chinese EQ-5D-5L study¹⁸ where the same cTTO tasks were conducted in a general population sample for similar health states, except that participants were told to imagine themselves, rather than a 10-year-old experiencing the hypothetical EQ-5D health states.

We examined the distribution of the cTTO data using a histogram and reported the observed health state values. Next, we modeled the effects of health state characteristics on the disutility (ie, $1 - \text{cTTO value}$) using 2 model specifications (ie, main effects only and main effects plus a “All 3” term [A3]). A3 takes the value of 1 for state “33333” and 0 for all other states. It was used to account for the extra utility decrement of state “33333.” For each model specification, we estimated 3 different models: (1) ordinary least square (OLS) as the base model, (2) a heteroscedastic model, and (3) a random-effects intercept model. We evaluated the models using 3 criteria: (1) monotonicity and logical consistency of the predictions, (2) significance and uncertainty of the estimated coefficients, and (3) prediction accuracy, measured using the mean absolute errors derived from the leave-one-state-out cross-validation analysis.¹⁹

Results

The demographic characteristics of the participants are presented in Table 1. In total, 418 participants completed the cTTO interviews. The mean age of the participants was 42.40 years (SD 14.62), with the majority being female (51.67%) and urban residents (64.35%) and completed either high school or higher education (66.91%). Our sample was slightly over-represented by females and those who completed higher education compared with the general population of China.¹⁶

TTO Task Performance

On average, participants took 35.70 minutes (SD 12.42) to complete the valuation interview and evaluated a mean of 13.21

Table 1. Demographic characteristics of the sample, N = 418.

Characteristics		n	%	General population,* %
Age group, year	18-29	108	25.84	25.70
	30-39	82	19.62	20.42
	40-49	93	22.25	21.85
	50-59	61	14.59	15.19
	> 60	74	17.7	16.84
Sex	Male	216	48.33	51.19
	Female	202	51.67	48.81
Hukou	Urban	269	64.35	66.67
	Rural	149	35.65	33.33
Education	Primary	138	33.01	33.75
	Junior high	166	39.71	41.70
	Senior high	59	14.11	15.02
	University and above	55	13.16	9.53
Health insurance	Urban employee	161	38.52	
	Residents	215	51.44	
	Commercial	12	2.87	
	Others	12	2.87	
	No	18	4.31	
Number of children	0	109	26.08	
	1	185	44.26	
	2	96	22.97	
	3	21	5.02	
	> 4	7	1.68	

*Based on the sixth National Population Census.¹⁶

(SD 9.00) pairs of alternative lives to reach the indifference point in a single cTTO task. The corresponding statistics for the Chinese EQ-5D-5L study¹⁸ were 31.57 minutes (SD 8.46) and 7.89 pairs (SD 2.55) of alternative lives. There were 74.16%, 59.33%, and 11.48% of the participants indicated that it was easy to understand the cTTO tasks, easy to differentiate between the health states, and difficult to determine the indifference point when completing the task, respectively. Comparatively, the corresponding percentages were 40.09%, 29.42%, and 12.83%, respectively, in the EQ-5D-5L valuation study of China.¹⁸ When the responses to the 5-point scale (1 = “agree” and 5 = “disagree”) were coded into a continuous score, the mean responses for the 3 feedback statements were 1.42 (SD 0.84), 1.65 (SD 0.96), and 3.45 (SD 1.35). The corresponding mean responses for the EQ-5D-5L valuation were 1.81 (SD 0.85), 3.12 (SD 1.24), and 2.08 (SD 0.98), respectively.

The data distribution of the cTTO values is presented in Figure 1. Data density was highest between the range of 0.5 and 1, with 21.89% being negative values (ie, < 0). Less than 2% of the values were at −1. The highest and lowest mean cTTO values were 0.924 (standard error [SE] 0.011) for state “11112” and −0.088 (SE 0.025) for state “33333.” State “33333” was the only state with a negative mean value, representing a large drop from the second-worst state “33323” (0.220; SE 0.040). The observed values decreased along the level sum score as shown in Appendix 1 found at <https://doi.org/10.1016/j.jval.2023.03.003>. The observed values for all health states are presented in Appendix 2 found at <https://dx.doi.org/10.1016/j.jval.2023.03.003>.

Modeling

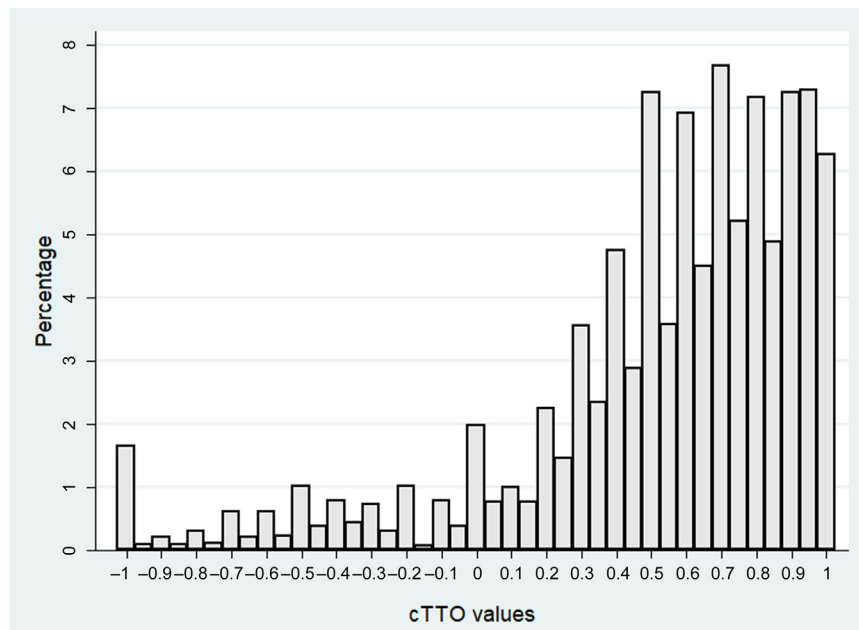
Table 2 summarizes the 3 models using the “main effects + A3” model specification, which outperformed their respective “main effects” only counterparts in prediction accuracy (see Appendix 3 found at <https://doi.org/10.1016/j.jval.2023.03.003> for the results of the “main effects” models). Overall, all models using the “main

effects + A3” specification achieved monotonicity, but one coefficient (level 2 usual activities, $P = .075$) in the OLS model was not significant at the 0.05 level. The prediction accuracy of the models was similar, with the random-effects model exhibiting the lowest cross-validation mean absolute error. In terms of model coefficients’ SEs, the random-effects model outperformed the OLS and heteroscedastic models.

Discussion

Our study demonstrated that it is feasible for members of the general public to value EQ-5D-Y-3L health states using the cTTO method. The data distribution was largely smooth without strong signs of data clustering or gaps. Most importantly, using the Chinese EQ-5D-5L valuation study¹⁸ as a reference, the EQ-5D-Y-3L cTTO tasks showed higher feasibility in terms of self-reported difficulties in task completion by participants. The interview time was slightly longer for the EQ-5D-Y-3L valuation tasks, which could be attributed to the higher QC standards implemented in the study.^{1,17} One QC indicator was whether the time for completing a cTTO task passed a minimum threshold, which the interviewers were aware of, and therefore possibly deliberately slowed down the interview pace by giving respondents more time to think or prompted them more carefully. In comparison, no such QC process was implemented in the adult EQ-5D-5L valuation study. Another benefit of implementing a QC process is that the interviewer effect was minimized in this study, given that all interviewers had similar data distributions (see Appendix 4 found at <https://doi.org/10.1016/j.jval.2023.03.003>). Furthermore, we found only one nontrader among our participants, further suggesting the suitability of using the cTTO method to value EQ-5D-Y-3L health states.

In this study, approximately 80% of the cTTO values elicited were positive, with the majority higher than 0.5; data clustering

Figure 1. Histogram of cTTO values.

cTTO indicates composite time trade-off.

at -1 and 0 was less salient compared with the EQ-5D-5L cTTO values in China. Moreover, all models produced reasonable modeling coefficients, which allows for the rejection of the hypothesis that high TTO values would fail in differentiating the relative importance of different health dimensions. Clearly, the utility decrements attached to EQ-5D-Y-3L were smaller compared with the published EQ-5D-3L value set of China.²⁰ Overall,

coefficients were smaller for all EQ-5D-Y-3L health problems except for “extreme” pain or discomfort (EQ-5D-Y-3L: 0.259 vs EQ-5D-3L: 0.243). Taking the coefficients for level-2 health problems as an example, the coefficients for mobility, looking after myself, usual activities, having pain or discomfort, and feeling worried or sad or unhappy were 0.047, 0.037, 0.030, 0.055, and 0.039 for EQ-5D-Y-3L, respectively, and 0.099, 0.103, 0.075, 0.093, and 0.088 for

Table 2. Model coefficients and prediction performance.

Variables/Models	OLS	Heteroscedastic	Random effects
Model coefficients, β , SE			
MO2	0.055, 0.014	0.047, 0.010	0.047, 0.011
MO3	0.145, 0.014	0.137, 0.013	0.147, 0.011
SC2	0.030, 0.014	0.038, 0.011	0.037, 0.011
SC3	0.124, 0.014	0.124, 0.011	0.122, 0.011
UA2	0.025*, 0.014	0.035, 0.010	0.030, 0.011
UA3	0.146, 0.015	0.147, 0.013	0.147, 0.011
PD2	0.070, 0.014	0.071, 0.009	0.055, 0.011
PD3	0.263, 0.014	0.269, 0.012	0.259, 0.011
AD2	0.033, 0.014	0.038, 0.010	0.039, 0.011
AD3	0.173, 0.014	0.168, 0.012	0.173, 0.011
A3	0.192, 0.023	0.200, 0.030	0.191, 0.017
Constant	0.046, 0.015	0.043, 0.010	0.050, 0.016
Model performance			
MAE	0.040	0.038	0.037
AIC	2743.66	1638.13	NA
BIC	2819.72	1790.25	NA

A3 indicates All 3; AD, anxiety or depression; AIC, Akaike information criterion; BIC, Bayesian information criterion; MAE, mean absolute error; MO, mobility; OLS, ordinary least square; PD, pain or discomfort; SC, self-care; SE, standard error; UA, usual activities.

*Coefficient UA2 was not significant at $P < .05$.

EQ-5D-3L. It should be noted that jointly using DCE data might not increase the utility decrements given that the scale range is mainly determined by the cTTO data.⁶ A possible explanation of the higher EQ-5D-Y-3L values observed is that the general public values children's life more than adults' life. The implication of this is that lifesaving interventions for children are more likely to be deemed cost-effective when EQ-5D-Y-3L values are used to estimate QALYs in economic evaluations.

Both TTO and DCE methods require respondents to make trade-offs and choices between alternatives, which mimics decision making in reality. DCE tasks ask respondents to trade between different health dimensions. Comparatively, the TTO method has the advantage of reflecting the QALY concept, given that it forces trade-off between quantity and quality of life. Moreover, the TTO method is more aligned with real-world medical decision making where survival and quality of life are the 2 primary outcomes.²¹ Thus far, all the published EQ-5D-Y-3L valuation studies^{7-9,22} have exhibited logically consistent and reasonable cTTO distributions, suggesting that cTTO-only value sets are possible if sufficient health states had been valued in those studies. For example, the Japanese study valued 26 states (18 from an orthogonal array + 8 handpicked states) using the cTTO method and estimated 3 cTTO models with consistent and significant coefficients.⁷ Kreimeier et al^{23,24} valued 17 states used in previous EQ-5D-3L valuation studies. Those data should also be sufficient for estimating a value set.³ More importantly, estimating a cTTO-only value set avoids the complex modeling analysis that is needed when both cTTO and DCE data are used. Therefore, cTTO-only value sets could be considered as an option in future EQ-5D-Y-3L valuation studies when the cTTO data are sufficient.

Conclusion

It is feasible to establish an EQ-5D-Y-3L value set using cTTO data alone. Given that the best strategy for combining the cTTO and DCE data to determine health state utilities is still unknown, using only cTTO data to establish EQ-5D-Y-3L value sets may be an option for future valuation studies.

Supplemental Material

Supplementary data associated with this article can be found in the online version at <https://doi.org/10.1016/j.jval.2023.03.003>.

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